APEX FX TRADING BOT

Product Requirements Document

Volume III — Frontend & Paper/Demo Trading Visibility

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Volume	III — Frontend, Paper Trading, Demo Trading Visibility
Builds On	PRD Vol I (v1.0) + PRD Vol II (v2.0)
Date	April 2026
Status	NEW — Frontend & Observability Layer
Classification	CONFIDENTIAL — Internal Use Only

PART A — PRD COMPLIANCE AUDIT: Was the PRD Followed?

This section is a formal cross-reference of what PRD Volumes I and II specified against what the README.md and SYSTEM_HIGHLIGHTS.md confirm was actually built. Every item is given a compliance rating.

1. PRD Volume I Compliance Assessment

1.1 Core Architecture & Technology Stack

PRD Vol I Requirement	Status	Notes
Python 3.11+ with full type hints	✓ BUILT	Confirmed in README stack
MetaTrader 5 via Python MT5 API	✓ BUILT	mt5_connector.py confirmed
6 per-instrument strategy modules	✓ BUILT	All 6 in strategies/engine.py
Regime Detection Engine (RDE)	✓ BUILT	TRENDING/RANGING/BREAKOUT/AVOID
ATR-based position sizing	✓ BUILT	Per-instrument profiles in risk/manager.py
Correlation manager	✓ BUILT	EUR+CHF, GBP+EUR, CAD+Gold caps
Economic calendar integration	✓ BUILT	calendar_feed.py
WTI crude oil feed for USD/CAD	✓ BUILT	commodity_feed.py
Order router with retry logic	✓ BUILT	order_router.py (3 attempts — Vol II requires 4)
Telegram alerts	✓ BUILT	telegram_alerts.py
Flask API + web dashboard	✓ BUILT	src/api/main.py + templates/index.html
SQLite database (5 core tables)	✓ BUILT	accounts, trades, signals, prices, settings
Structured JSON logging	✓ BUILT	logger.py
Health heartbeat monitor	✓ BUILT	health_monitor.py
APScheduler for scheduled jobs	✓ BUILT	Confirmed in README job table
Per-instrument daily loss limits	⚠ PARTIAL	Master Prompt Gap #5: enforcement logic unclear
Trailing stops activated in main loop	⚠ PARTIAL	Master Prompt Gap #1: methods exist, not called
4-attempt retry with exact ms delays	✗ GAP	Only 3 attempts — Vol II spec is 4
Max hold time / weekend policy	✗ GAP	Master Prompt Gaps #7 & #8
BoJ intervention flag persistence (restart)	✗ GAP	Minor gap from Master Prompt audit

1.2 Risk Management Compliance

PRD Vol I Risk Requirement	Status	Verified Via
Per-pair risk % (0.75%–1.5%)	✓ BUILT	risk_config.json per SYSTEM_HIGHLIGHTS table
ATR multiplier per pair (1.5×–2.0×)	✓ BUILT	Per-instrument profiles confirmed
Daily circuit breaker 8%	✓ BUILT	risk/manager.py
Monthly drawdown 15%	✓ BUILT	Confirmed in SYSTEM_HIGHLIGHTS
Max lot size per pair	✓ BUILT	Risk config profiles
EUR/USD+USD/CHF correlation block	✓ BUILT	correlation.py
GBP/EUR combined cap	✓ BUILT	correlation.py
Dynamic pip value from MT5 (not hardcoded)	⚠ PARTIAL	Master Prompt Gap #3: formula discrepancy
News blackout 30min/45min gold	✓ BUILT	calendar_feed.py gold-specific
Session gates per instrument	✓ BUILT	All 6 strategies
Spread filter pre-trade	✓ BUILT	order_router.py
Spread-to-ATR ratio filter (< 0.15)	✗ GAP	Master Prompt Gap #4: missing entirely

1.3 PRD Volume II Compliance Assessment

PRD Vol II Module	Status	Location
Portfolio Heat Monitor	✓ BUILT	risk/portfolio_heat_monitor.py
Volatility Scaler (VRS)	✓ BUILT	risk/volatility_scaler.py
Signal Scorer (0-100 weighted)	✓ BUILT	risk/signal_scorer.py
Kelly Criterion Sizer	✓ BUILT	risk/kelly_sizer.py
Fill Analyzer + BQS	✓ BUILT	risk/fill_analyzer.py
Regime Drift Detector (RDD)	✓ BUILT	risk/regime_drift_detector.py
Performance Analyzer (self-learning)	✓ BUILT	analysis/performance_analyzer.py
COT Parser (CFTC integration)	✓ BUILT	data/cot_parser.py
Time Exit Manager	✓ BUILT	execution/time_exit_manager.py
Calendar Filter (day-of-week, month-end)	✓ BUILT	execution/calendar_filter.py
Swap Filter (rollover protection)	✓ BUILT	data/swap_filter.py
4-tier Telegram alert system	✓ BUILT	INFO/WARNING/CRITICAL/EMERGENCY
7 new database tables (Vol II)	✓ BUILT	fill_quality, signal_scores, kelly_history, etc.
API v2 endpoints (8 new routes)	✓ BUILT	/api/v2/heat, /vrs, /bqs, /kelly, /rdd, /cot, etc.

PRD Vol II Module	Status	Location
Weekly performance report	✓ BUILT	weekly_reports table + scheduled job
133 tests passing	✓ BUILT	pytest test suite confirmed
Spread percentile filter (80th pct)	⚠ PARTIAL	Referenced but dynamic 20-day window unclear
Secondary broker account specification	✗ NOT BUILT	Vol II 21.2 — infrastructure spec only
Parameter change governance (15% rule)	⚠ PARTIAL	Performance analyzer built; governance gate unclear
XAU Defensive Mode (6h minimum)	⚠ PARTIAL	VRS defensive mode exists; 6h hold unconfirmed

1.4 Overall PRD Compliance Verdict

✓ STRONG COMPLIANCE — Both PRDs Were Largely Followed

Volume I: ~78% fully compliant. 4 partial gaps, 3 missing items.Volume II: ~86% fully compliant. 4 partial, 1 not built (secondary broker — infrastructure only).The 8 critical gaps identified in the Master Coding Prompt are the most important outstanding items from Volume I. The 11 Volume II modules are all confirmed built per README/SYSTEM_HIGHLIGHTS.Conclusion: The system is substantively complete and production-capable. The gaps do not prevent demo trading — they are refinements that must be resolved before scaling to live capital.

1.5 Is a Volume III Needed?

YES. There is one major category that both PRD Volumes I and II deliberately deferred:

The Critical Gap Not Covered by Volumes I or II

Neither volume specified what the user sees, how paper trading is monitored in real time, how the demo trading session on MT5 is observed and evaluated, or what the frontend dashboard must actually contain beyond 'a web dashboard exists.' The current index.html is unspecified — it could be anything.Volume III exists for one purpose: to define, in complete detail, the frontend user interface, the paper trading observation layer, and the demo trading visibility system — everything the operator needs to watch, understand, and trust the bot before risking real capital.

PART B — PRD VOLUME III: Frontend & Trading Visibility Specification

2. Executive Summary — Volume III

PRD Volume III defines the complete user-facing layer of APEX FX. It covers three interconnected concerns:

- The Dashboard — the real-time web UI the operator uses to monitor the bot in all modes
- The Paper Trading Observation System — a sandboxed simulation environment where signals fire and virtual trades are logged without any MT5 connection, allowing full strategy validation in any environment
- The Demo Trading Visibility System — a live MT5-connected mode where the bot trades on a demo broker account with full observability: every trade, every signal decision, every risk calculation is visible in real time

The frontend is not a cosmetic addition. It is the operator's primary tool for building confidence in the system before committing real capital, diagnosing problems, and monitoring the system in production.

3. Frontend Architecture Overview

3.1 Technology Stack — Frontend

Component	Technology	Rationale
Frontend Framework	Vanilla HTML/CSS/JavaScript (no build step)	MT5 runs on Windows VPS — no Node.js bundler guaranteed
UI Style	Custom CSS with CSS variables — dark trading theme	Low latency; no CDN dependency risk
Real-Time Data	Server-Sent Events (SSE) via Flask /api/stream	One-way push from server; no WebSocket complexity
Charts	Lightweight-charts (TradingView) via CDN	Professional OHLC candle charts; free
Tables	Custom HTML tables — sortable, filterable	No dependency; fully controllable
State Management	localStorage + in-memory JS object	Persists user preferences (theme, symbol filters)
API Communication	Fetch API with retry + error boundary	Native browser; no axios dependency

Component	Technology	Rationale
Notifications	Browser Notification API + in-page toast system	Redundant with Telegram

3.2 Trading Modes & Dashboard States

The dashboard must adapt to three operating modes. The active mode is declared in config.yaml (system.mode) and displayed permanently in the header:

Mode	MT5 Connected?	Real Orders?	Dashboard Indicator
PAPER	NO — purely simulated	NO — virtual P&L only	Blue banner: PAPER TRADING MODE
DEMO	YES — demo broker account	YES — real execution, fake money	Orange banner: DEMO TRADING — MT5 CONNECTED
LIVE	YES — live broker account	YES — real capital at risk	Red banner: LIVE TRADING — REAL CAPITAL

4. Dashboard Layout Specification

4.1 Page Structure

Dashboard Layout (7 Zones)

ZONE 1 — Top Header Bar (fixed, always visible)ZONE 2 — System Status Strip (live KPIs in one row)ZONE 3 — Left Sidebar (symbol watchlist + mode controls)ZONE 4 — Main Panel (tabbed content area — see Section 4.2)ZONE 5 — Right Sidebar (active signals queue + risk gauges)ZONE 6 — Positions Panel (open trades table — slides up from bottom on trade open)ZONE 7 — Event Log (collapsible real-time feed of all bot decisions)

4.2 Zone 1 — Top Header Bar

Fixed header, always visible regardless of scroll. Contains:

- APEX FX logo/wordmark (left)
- Mode indicator pill: PAPER (blue) / DEMO (orange) / LIVE (red) with pulsing dot
- MT5 connection status: green dot + 'Connected' or red dot + 'Disconnected' + reconnect button
- Account equity display: real-time equity (large), balance (small below), unrealized P&L (colored green/red)
- VRS Scalar badge: shows current VRS value (0.40–1.00) with color: green ≥0.85, amber 0.70–0.84, red <0.70
- Portfolio Heat gauge: mini donut chart showing heat % with color coding (green <2%, amber 2–3%, red >3%)
- Clock: current UTC time + local time (since all bot logic uses UTC)

- Settings gear icon (right) — opens settings modal

4.3 Zone 2 — System Status Strip

Single row of 8 KPI tiles immediately below the header. Each tile shows a metric, a value, and a trend arrow:

Tile #	Metric	Data Source	Color Logic
1	Today's P&L (%)	/api/stats daily_pnl	Green if positive, red if negative
2	Open Positions	/api/positions count	White — informational
3	Win Rate (30d)	/api/stats win_rate_30d	Green >52%, amber 45-52%, red <45%
4	Profit Factor (30d)	/api/stats profit_factor_30d	Green >1.5, amber 1.2-1.5, red <1.2
5	Portfolio Heat %	/api/v2/heat heat_pct	Green <2%, amber 2-3%, red >3%
6	BQS (Broker Quality)	/api/v2/bqs score	Green >80, amber 65-80, red <65
7	RDD Status (worst pair)	/api/v2/rdd worst_status	GREEN/AMBER/RED/CRITICAL
8	VRS Scalar	/api/v2/vrs scalar	Green 1.00, amber 0.70, red 0.40-0.50

4.4 Zone 3 — Left Sidebar: Watchlist & Controls

Collapsible left panel (280px wide). Contains two sections:

Symbol Watchlist

- 6 rows — one per instrument (EUR/USD, GBP/USD, USD/JPY, USD/CHF, USD/CAD, XAU/USD)
- Each row shows: symbol name, current bid/ask, daily change pip/%, mini sparkline (1h), current regime badge (T=Trending, R=Ranging, B=Breakout, ⊘=Avoid)
- Click any symbol to load it in the main chart panel
- Each symbol row has a color-coded left border: green (active, no issues), amber (RDD AMBER), red (RDD RED / suspended)

Mode Control Panel

- Active Mode display (read from config)
- 'Start Scanning' / 'Pause Scanning' toggle button
- 'Emergency Stop All' button — closes all positions immediately (red, requires double-click confirmation)
- 'Run Manual Signal Scan' button — triggers immediate scan outside the scheduler
- Heartbeat status indicator: shows seconds since last heartbeat

4.5 Zone 4 — Main Panel (Tabbed)

The central content area uses 6 tabs:

Tab 1: CHART

- Full-width TradingView Lightweight Charts OHLC candlestick chart for the selected symbol
- Timeframe selector: M15 / H1 / H4 / D1
- Overlays auto-drawn from /api/market/indicators: 20 EMA (blue), 50 EMA (orange), 200 EMA (red), Bollinger Bands (grey)
- All open positions shown as horizontal lines: entry price (white dashed), stop-loss (red solid), take-profit (green solid)
- All recent signals shown as arrows on the chart: green up-arrow for buy signal (even if rejected), red down-arrow for sell signal, grey for rejected signals
- Signal popup on click: shows the full signal score breakdown (Trend Alignment, Momentum, Regime Quality, Session/Timing, Execution Quality — all 5 factors with scores)
- Regime zone shading: background color changes subtly based on current regime (blue-tint for TRENDING, grey for RANGING, yellow for BREAKOUT_PENDING, red-tint for AVOID)

Tab 2: SIGNALS

Signal history table. All signals generated in the last 7 days. Columns:

- Timestamp (UTC), Symbol, Direction (BUY/SELL), Signal Score (0-100 with color grade), Action Taken (OPENED / REJECTED + reason), Entry Price, SL, TP, Current P&L (if open), Outcome (WIN/LOSS/OPEN)
- Filter bar: by symbol, by date range, by outcome, by minimum score
- Score grade color coding: <60 grey (REJECT), 60-74 amber (MARGINAL), 75-84 white (STANDARD), 85-100 gold (HIGH CONVICTION)
- Clicking any row expands the full signal score breakdown: all 5 factor scores, COT index at signal time, VRS scalar at signal time, regime at signal time

Tab 3: PERFORMANCE

Analytics dashboard. Four sub-sections:

- P&L Curve: line chart of daily cumulative P&L for the last 90 days. Overlay: rolling drawdown from equity high (shaded red area below zero). Toggle between % and absolute $
- Per-Pair Stats table: for each of the 6 pairs — total trades, win rate, avg R:R achieved, profit factor, avg signal score of wins vs losses, current Kelly risk %, current RDD status
- Score vs Outcome chart: scatter plot — X-axis = signal score (0-100), Y-axis = trade outcome in R multiples. Shows whether higher scores produce better outcomes. Color by pair.
- Session Performance heatmap: 24×5 grid (hours × weekdays). Each cell is colored by win rate for trades opened in that hour/day combination. Reveals the best and worst times for each strategy.

Tab 4: RISK DASHBOARD

Real-time risk visualization. Four panels:

- Portfolio Heat Gauge: large circular gauge (0–6%). Current heat shown with needle. Bands: Cold (0-2% green), Warm (2-3% amber), Hot (3-4% red), Critical (>4% dark red). Below: breakdown by pair showing each pair's contribution to total heat.

- VRS History chart: area chart of VRS scalar over the last 30 days. Shows how often the system operated below full size. Overlay: XAU Defensive Mode periods (shaded).
- Kelly Risk Table: for each pair — current Kelly fraction, min floor, max cap, consecutive losses (with cold streak ladder status), activation threshold progress (X of 30 trades)
- Correlation Exposure matrix: 6×6 grid showing current cross-pair correlation. Cells that are at the correlation limit flash amber. Cells that have triggered a block show red.

Tab 5: EXECUTION

Broker quality and execution monitoring. Three panels:

- BQS Trend chart: Broker Quality Score over the last 12 weeks. Threshold line at 65 (CRITICAL) and 80 (target). Components breakdown below: execution shortfall, requote rate, fill latency, spread expansion — each as its own metric tile with 4-week trend sparkline.
- Fill Quality Table: last 100 fills with columns — timestamp, symbol, expected price, actual fill, shortfall (pips), spread at submission, spread at fill, latency (ms), requote? (Y/N)
- MISSED_SIGNAL log: all signals that reached order submission but failed after 4 attempts. Shows symbol, time, reason, and which attempt number failed last.

Tab 6: WEEKLY REPORT

Structured weekly report viewer. Displays the most recent weekly_report_YYYYMMDD.json in a formatted, human-readable layout matching the structure defined in PRD Vol II Section 22.2:

- Executive Summary section
- Instrument breakdown section (per-pair performance table)
- Execution quality section
- Risk snapshot section
- RDD & Kelly status section
- COT update section (COT Index per pair, change from prior week)
- Next week outlook (high-impact events + pre-declared adjustments)
- Archive selector: dropdown to view any prior week's report from the database

4.6 Zone 5 — Right Sidebar: Active Signals & Risk Gauges

Collapsible right panel (260px wide). Always-live feed showing:

- Active Signal Queue: signals that have passed the RDE and are currently in the pre-trade check pipeline. Each shows the symbol, direction, current score (live-updating as each check runs), and which check is currently processing (displayed as a 15-step progress indicator).
- Open Positions mini-list: compact view of all open trades with entry, current price, floating P&L, and time open.
- News Alert Feed: upcoming high-impact events in the next 4 hours. Each event shows: time (UTC countdown), currency affected, event name, impact level (red/orange/yellow), and which pairs will be blocked.

4.7 Zone 6 — Positions Panel

A persistent panel that appears at the bottom of the screen whenever there are open positions. Slides up to a configurable height (default: 200px). Contains:

- Full positions table: all open positions with all columns — ticket #, symbol, direction, lots, open time, entry price, current price, stop-loss, take-profit, floating P&L ($), floating P&L (%), time open, signal score at open, trailing stop active? (Y/N)
- Actions per row: 'Close Position' button (requires single-click confirmation), 'Modify SL/TP' button (inline edit)
- Footer row: Total floating P&L across all positions, total lots, effective portfolio heat

4.8 Zone 7 — Event Log

A collapsible bottom strip (default: collapsed). When expanded shows a real-time, auto-scrolling log of every decision the bot makes. This is the 'bot's inner monologue' — critical for understanding and debugging. Each log entry shows:

- Timestamp (UTC, millisecond precision)
- Symbol (color-coded)
- Event type badge: SIGNAL_DETECTED / CHECK_PASSED / CHECK_FAILED / ORDER_PLACED / ORDER_MODIFIED / ORDER_CLOSED / CIRCUIT_BREAKER / RDD_UPDATE / HEARTBEAT
- Details: human-readable explanation (e.g. 'EUR/USD SELL signal scored 72 — MARGINAL. Spread-to-ATR check passed (0.09). Portfolio heat check BLOCKED — heat at 3.2% (HOT level). Signal rejected.')
- Filter: by symbol, by event type, by severity

5. Paper Trading System Specification

5. Paper Trading System — Full Specification

5.1 What Paper Trading Is in This System

Paper trading mode (system.mode = PAPER) is a full strategy simulation that runs without any MT5 connection. The strategy engine, regime detector, signal scorer, risk engine, and all 11 Volume II modules run exactly as they would in live trading — but instead of calling the MT5 order router, they call a Paper Trading Engine that maintains a virtual account, virtual positions, and virtual P&L in the SQLite database.

Paper Trading Design Principle

Paper trading must be a 100% faithful simulation of the live system. Every pre-trade check, every signal score, every risk filter runs identically. The ONLY difference is where price data comes from (historical or delayed feed instead of live tick) and where orders go (virtual account instead of MT5). If paper trading is materially different from live trading, it provides false confidence.

5.2 Paper Trading Engine — New Module: src/paper/paper_engine.py

This new module must be built for Volume III. It is NOT a simplified mock. It is a full virtual execution system.

Virtual Account Manager

- Maintains a virtual account with configurable starting balance (default: $10,000)
- Tracks virtual equity = starting balance + all closed trade P&L + all open floating P&L
- Tracks virtual margin based on simulated leverage (default: 1:100)
- Persists all virtual account state to a paper_account table in SQLite
- Can be reset to starting balance via the dashboard without affecting real trade history

Virtual Order Execution

- When the strategy engine generates a signal that passes all pre-trade checks, the paper engine creates a virtual position
- Fill price simulation: uses the current market bid/ask from the MT5 data feed (even in paper mode, OHLC data is fetched from MT5 in read-only mode) or from a simulated slippage model if MT5 is unavailable
- Slippage simulation: adds a realistic slippage to the fill price based on the instrument's historical average slippage (configurable per symbol in paper_config.yaml)
- Spread simulation: applies the current live spread (or a configurable average spread if no live feed) to opening positions

- Commission simulation: deducts a configurable commission per lot at open and at close

Virtual Position Management

- All virtual positions are stored in paper_positions table
- Stop-loss and take-profit are monitored against live OHLC data on every H1 bar close
- If H1 low breaches the stop-loss price, the position is closed at the stop price (simulating worst-case realistic fill)
- If H1 high reaches the take-profit, the position is closed at the TP price
- Trailing stops are activated and updated exactly as the live system would
- Time exit manager rules apply identically: max hold times, forward progress rule, weekend policy all enforced

5.3 Paper Trading Database Schema

```
CREATE TABLE IF NOT EXISTS paper_account (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  timestamp TEXT NOT NULL,
  balance REAL NOT NULL,          -- Running virtual balance
  equity REAL NOT NULL,           -- Balance + floating P&L
  floating_pnl REAL DEFAULT 0,
  realized_pnl REAL DEFAULT 0,
  total_trades INTEGER DEFAULT 0,
  reset_count INTEGER DEFAULT 0   -- How many times paper account was reset
);

CREATE TABLE IF NOT EXISTS paper_positions (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  ticket TEXT NOT NULL UNIQUE,    -- Paper ticket (format: PAPER-YYYYMMDD-NNN)
  symbol TEXT NOT NULL,
  direction TEXT NOT NULL,        -- BUY or SELL
  lot_size REAL NOT NULL,
  open_time TEXT NOT NULL,
  open_price REAL NOT NULL,
  simulated_spread REAL,
  simulated_commission REAL,
  stop_loss REAL NOT NULL,
  take_profit REAL NOT NULL,
  trailing_stop_active INTEGER DEFAULT 0,
  trailing_stop_price REAL,
  signal_score REAL,
  kelly_risk_pct REAL,
  vrs_scalar REAL,
  regime TEXT,
  current_price REAL,
  floating_pnl REAL DEFAULT 0,
  status TEXT DEFAULT 'OPEN',     -- OPEN, CLOSED_SL, CLOSED_TP, CLOSED_TIME, CLOSED_MANUAL
  close_time TEXT,
  close_price REAL,
  realized_pnl REAL
);

CREATE TABLE IF NOT EXISTS paper_trades_log (
  -- Same schema as trades table but with paper_ prefix source
  -- Full audit trail of every paper trade open and close
);
```

5.4 Paper Trading Dashboard Flow

When the system is in PAPER mode, the dashboard behaves as follows:

- Blue 'PAPER TRADING MODE' banner is permanently displayed in the header
- Zone 2 status strip shows paper account equity, paper daily P&L, paper win rate — all clearly labeled as paper statistics
- The Positions Panel (Zone 6) shows paper positions, labeled '[PAPER]' with a blue badge on each row
- The Chart tab shows paper signals as arrows (green dashed for paper buy, orange dashed for paper sell)
- Zone 7 Event Log shows paper trade events with [PAPER] prefix on every event
- A 'Reset Paper Account' button appears in the left sidebar control panel — resets virtual balance to configured starting amount, archives existing paper trade history with a reset timestamp
- The Performance tab shows two sub-modes: 'Paper Stats' and 'Combined Stats' — allowing comparison of paper performance against any live trade history that exists

5.5 Paper Trading Without MT5 (Fully Offline Mode)

The paper trading engine must be able to run with NO MT5 connection at all. This supports development, testing, and environments where MT5 cannot be installed. In this sub-mode:

- Price data is sourced from the last available OHLC data in the SQLite prices table, or from a configurable CSV file of historical prices
- The system replays candle by candle in accelerated time (configurable: 1× to 1000× speed)
- All strategy logic runs against the historical data exactly as it would in live trading
- This enables backtesting through the live dashboard — not just through scripts — making the bot's behavior fully observable in the UI

6. Demo Trading Visibility System

6. Demo Trading Visibility — Full Specification

6.1 What Demo Trading Mode Is

Demo trading (system.mode = DEMO) means the bot is fully connected to a MetaTrader 5 demo broker account. Real order routing happens, real MT5 execution occurs, but no real money is at risk. This mode is the final validation step before live capital deployment.

The observability requirement for demo mode is higher than paper mode because in demo mode the operator must be able to verify that every component of the live system is functioning correctly — order routing, fills, trailing stops, circuit breakers, all 15 pre-trade checks. Any discrepancy between what the bot decides and what the dashboard shows represents a monitoring failure.

6.2 Demo Mode Dashboard Differences

Feature	Paper Mode	Demo Mode
Header banner color	Blue	Orange (MT5 DEMO CONNECTED)
Order execution	Virtual paper engine	Real MT5 order_send() calls
Position data source	paper_positions table	Live mt5.positions_get() on every refresh
P&L displayed	Simulated from OHLC data	Real MT5 floating P&L from broker
Fill price	Simulated with slippage model	Actual broker fill price from MT5
Latency display	N/A (no network)	Real fill latency from fill_analyzer.py
BQS calculation	N/A	Live — every fill runs through fill_analyzer.py
Trailing stop updates	Simulated	Real MT5 order_modify() calls
Account equity	Virtual paper balance	Real MT5 account_info().equity
Signal arrows on chart	Dashed lines	Solid lines (real signal decisions)

6.3 Demo Trade Flow Visibility — The 15-Step Pipeline

Every signal that fires in demo mode must be visible to the operator in real time through the Event Log (Zone 7) and the Right Sidebar Signal Queue (Zone 5). The operator must be able to see exactly where in the 15-step pre-trade pipeline a signal is, why it was approved or rejected, and what final position size was calculated.

The Event Log entry format for each step of a passing signal:

```
[14:32:01.445 UTC] EURUSD — SIGNAL_DETECTED — Direction: BUY — Raw Score: 67 — Regime:
TRENDING
[14:32:01.446 UTC] EURUSD — CHECK 1 PASSED — RDE: TRENDING (ADX=31.2, above threshold 25)
[14:32:01.447 UTC] EURUSD — CHECK 2 PASSED — Session Gate: London-NY overlap (14:32 UTC)
[14:32:01.448 UTC] EURUSD — CHECK 3 PASSED — News Blackout: no high-impact events within
30min
[14:32:01.449 UTC] EURUSD — CHECK 4 PASSED — Spread Percentile: 42nd pct (threshold: <80th)
[14:32:01.450 UTC] EURUSD — CHECK 5 PASSED — Spread/ATR Ratio: 0.08 (threshold: <0.15)
[14:32:01.451 UTC] EURUSD — CHECK 6 PASSED — Swap Filter: 7.4h until rollover
(short-duration threshold not triggered)
[14:32:01.452 UTC] EURUSD — CHECK 7 PASSED — COT Alignment: COT Index 61 (>55 = long
confirm) +10 pts added
[14:32:01.453 UTC] EURUSD — CHECK 8 PASSED — Signal Score: 77 (STANDARD — threshold 60) —
Modifier: ×1.00
[14:32:01.454 UTC] EURUSD — CHECK 9 PASSED — VRS Scalar: 1.00 (calm vol environment)
[14:32:01.455 UTC] EURUSD — CHECK 10 PASSED — Kelly Size: 1.35% (38 trades, Kelly active)
[14:32:01.456 UTC] EURUSD — CHECK 11 PASSED — Portfolio Heat: 1.8% (COLD — new positions
permitted)
[14:32:01.457 UTC] EURUSD — CHECK 12 PASSED — Correlation: EUR+CHF OK (no USD/CHF position
open)
[14:32:01.458 UTC] EURUSD — CHECK 13 PASSED — Open Positions: 2/4 (below max 4)
[14:32:01.459 UTC] EURUSD — CHECK 14 N/A — Order size 0.18 lots (below 1.0 lot split
threshold)
[14:32:01.521 UTC] EURUSD — ORDER_PLACED — Ticket: 123456 — BUY 0.18 lots @ 1.08921
                   SL: 1.08781 (14.0 pips / 1.5×ATR) — TP: 1.09154 (23.3 pips / 2.5×ATR)
                   Final Risk: 1.35% — Position Value: $18,231 — Latency: 62ms
```

And for a rejected signal:

```
[14:51:33.112 UTC] XAUUSD — SIGNAL_DETECTED — Direction: BUY — Raw Score: 55 — Regime:
RANGING
[14:51:33.113 UTC] XAUUSD — CHECK 1 PASSED — RDE: RANGING (ADX=18.1)
[14:51:33.114 UTC] XAUUSD — CHECK 2 PASSED — Session Gate: London session active
[14:51:33.115 UTC] XAUUSD — CHECK 3 PASSED — News Blackout: clear
[14:51:33.116 UTC] XAUUSD — CHECK 4 PASSED — Spread Percentile: 55th pct
[14:51:33.117 UTC] XAUUSD — CHECK 5 PASSED — Spread/ATR Ratio: 0.11
[14:51:33.118 UTC] XAUUSD — CHECK 6 SKIPPED — Swap Filter: >3h to rollover
[14:51:33.119 UTC] XAUUSD — CHECK 7 FAIL — COT Alignment: Gold COT Index 91 (≥90
CONTRAINDICATION) -15 pts
[14:51:33.120 UTC] XAUUSD — CHECK 8 REJECTED — Signal Score: 55 (below threshold 60) —
SIGNAL ABANDONED
                   Score breakdown: Trend=20, Momentum=12, Regime=8, Session=10,
Execution=5, COT=-15 → TOTAL=55
```

6.4 Demo Mode — Performance Validation View

The Performance tab in demo mode adds a critical comparison panel not present in paper mode:

- BACKTEST vs DEMO comparison: side-by-side table showing each KPI from the backtesting phase against the current demo trading results. Highlighted in amber if demo deviates more than 20% from backtest on any metric — this is the Vol I Section 7.3 '20% deviation' check made visual.
- A running counter shows how many demo trades have been completed per pair vs the minimum threshold for live deployment confidence (e.g. 'EUR/USD: 23/50 trades — insufficient for Kelly activation, 10/20 weeks demo minimum not yet reached')
- Live-vs-Expected slippage chart: plots actual demo fill prices against the expected entry price for every trade. Shows whether the broker is executing at the expected price.

6.5 The 'Ready for Live?' Checklist

A dedicated panel in the Performance tab called 'Live Deployment Readiness' shows a checklist of all conditions that must be met before transitioning from DEMO to LIVE mode:

Condition	Requirement	Current Status (example)
Demo runtime	≥ 8 weeks continuous	Week 3 of 8
Total demo trades	≥ 50 across all pairs	34/50
Demo win rate	≥ 45%	54% ✓
Demo profit factor	≥ 1.2	1.38 ✓
Demo max drawdown	< 15%	6.2% ✓
Demo vs backtest deviation	< 20% on all KPIs	11% max deviation ✓
BQS (demo broker quality)	≥ 72 for 4 consecutive weeks	Week 2 of 4 (BQS: 81)
All critical gaps fixed	0 CRITICAL items in audit	1 gap pending (trailing stop loop)
RDD status all pairs	All GREEN	4 GREEN, 2 AMBER (USD/CHF, USD/CAD)
VPS stability	0 crashes in 4 weeks	1 crash at Week 1 (resolved)

7. New Backend Modules Required for Volume III

7. Volume III — New Modules & API Endpoints

7.1 New Python Modules

Module	Location	Purpose
paper_engine.py	src/paper/paper_engine.py	Virtual order execution, position management, P&L simulation
paper_account.py	src/paper/paper_account.py	Virtual account state, balance tracking, reset logic
sse_streamer.py	src/api/sse_streamer.py	Server-Sent Events stream for real-time dashboard updates
event_log.py	src/monitoring/event_log.py	Structured event log for Zone 7 — all 15-step pipeline decisions
readiness_checker.py	src/analysis/readiness_checker.py	Live Deployment Readiness checklist evaluation
chart_data.py	src/api/chart_data.py	OHLC + indicator data formatted for Lightweight Charts

7.2 New API Endpoints — Volume III

Endpoint	Method	Returns
/api/stream	GET (SSE)	Server-Sent Events stream — account, positions, signals, heat, VRS every 2s
/api/v3/paper/account	GET	Current paper account state (balance, equity, P&L, trade count)
/api/v3/paper/positions	GET	All open paper positions with real-time floating P&L
/api/v3/paper/trades	GET	Closed paper trade history (paginated, filterable)
/api/v3/paper/reset	POST	Archive and reset paper account to starting balance
/api/v3/event-log	GET	Last N event log entries (filterable by symbol, type, severity)
/api/v3/chart-data/<symbol>/<tf>	GET	OHLC candles + indicator values formatted for Lightweight Charts
/api/v3/signal-detail/<signal_id>	GET	Full 15-step pipeline breakdown for any historical signal
/api/v3/readiness	GET	Live deployment readiness checklist with all condition statuses
/api/v3/comparison/backtest-vs-demo	GET	Side-by-side KPI comparison for all 6 pairs

Endpoint	Method	Returns
/api/v3/weekly-report/archive	GET	List of all available weekly report files
/api/v3/mode	GET	Current system mode (PAPER/DEMO/LIVE) + MT5 connection status

7.3 New Database Tables — Volume III

```
CREATE TABLE IF NOT EXISTS event_log (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  timestamp TEXT NOT NULL,
  symbol TEXT,
  event_type TEXT NOT NULL,   -- SIGNAL_DETECTED, CHECK_PASSED, CHECK_FAILED, ORDER_PLACED,
etc.
  check_step INTEGER,         -- 1-15 for pre-trade pipeline steps, NULL for other events
  severity TEXT DEFAULT 'INFO', -- INFO, WARNING, CRITICAL, EMERGENCY
  details TEXT NOT NULL,      -- Human-readable description
  signal_id INTEGER,          -- FK to signals table if related to a signal
  raw_data TEXT               -- JSON blob of all values at time of event
);

CREATE TABLE IF NOT EXISTS readiness_log (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  timestamp TEXT NOT NULL,
  condition_name TEXT NOT NULL,
  requirement TEXT NOT NULL,
  current_value TEXT NOT NULL,
  is_met INTEGER NOT NULL,    -- 0 or 1
  snapshot_week TEXT          -- ISO week identifier
);
```

PART C — VOLUME III MASTER CODING PROMPT

8. Volume III — Master AI Coding Prompt

Copy everything from the box below and give it verbatim to your AI coding assistant. Follow the session order at the bottom.

HOW TO USE THIS PROMPT

Paste the entire Section 8.1 prompt into a fresh AI session. Then implement one session at a time in the order given in Section 8.2. Do NOT skip sessions. Each session builds on the previous one. The frontend cannot be completed until the SSE streamer and event log backend are built.

8.1 The Prompt (Paste Verbatim to Your AI)

```
═══════════════════════════════════════════════════════════════
APEX FX TRADING BOT — VOLUME III IMPLEMENTATION PROMPT
Frontend, Paper Trading System, Demo Trading Visibility
═══════════════════════════════════════════════════════════════
```

PART 1 — YOUR ROLE

You are a senior full-stack engineer completing the APEX FX Trading Bot — a fully automated Python algorithmic trading system for MetaTrader 5. The backend is substantially complete (Volumes I and II). Your job is to build the user-facing layer: the real-time web dashboard, the paper trading engine, and the demo trading observability system.

You must produce code that integrates with the EXISTING backend without breaking it. Every new module you write must follow the same patterns as the existing code: structlog logging, threading.Lock() for shared state, timezone-aware UTC datetimes, config from YAML/JSON files, pytest tests for every module.

PART 2 — EXISTING SYSTEM CONTEXT

The existing backend (do not rewrite these):

- src/api/main.py — Flask application with all Vol I + II endpoints. You will ADD new endpoints here, not replace existing ones.
- src/data/database.py — SQLiteManager with 12 tables. ADD the 3 new Vol III tables as migration methods.
- src/monitoring/logger.py — structlog logger. USE this for all logging.
- src/risk/ — All risk modules (heat monitor, VRS, Kelly, signal scorer, fill analyzer, RDD). These are all operational. Your frontend must display their outputs, not duplicate them.

- templates/index.html — The existing dashboard HTML. You will REPLACE this file entirely with the new full-featured dashboard per the Vol III spec.

PART 3 — MANDATORY CODING RULES

- Python 3.11+ with full type hints on every function signature. Docstrings on every function.
- Frontend: HTML5, CSS3 with CSS variables, ES6+ JavaScript. No external build tools. No TypeScript. No React/Vue — plain JavaScript only.
- All JavaScript in a single app.js file (or inline in index.html for simplicity). No multiple JS file imports unless necessary.
- The Lightweight Charts library (TradingView) is loaded from CDN: https://unpkg.com/lightweight-charts@4.0.0/dist/lightweight-charts.standalone.production.js
- All real-time data comes via Server-Sent Events at GET /api/stream — never via polling. Polling is not permitted for live data.
- All configurable values (starting paper balance, chart colors, refresh intervals) live in a new vol3_config.yaml in the config/ directory.
- Never use print(). Use the existing structlog logger from src/monitoring/logger.py.
- All new API endpoints follow the existing pattern in src/api/main.py: Flask Blueprint, error handling via try/except returning JSON error responses, authentication if the existing API has any.
- After each module: write a pytest test file in tests/. Mock Flask and database — tests must run without a live MT5 terminal or browser.

PART 4 — MODULE SPECIFICATIONS

Build each module in the order given in Part 6. Full specifications are in PRD Volume III. Here are the critical implementation details for each:

MODULE 1: src/monitoring/event_log.py — EventLogger class

- Stores every pre-trade pipeline step as a structured event in the event_log database table
- Method: log_event(symbol, event_type, details, check_step=None, severity='INFO', signal_id=None, raw_data=None)
- Thread-safe: uses threading.Lock() around all database writes
- Method: get_events(symbol=None, event_type=None, severity=None, limit=200) → list of event dicts
- EventLogger is instantiated once in main.py and passed to: OrderRouter, SignalScorer, PortfolioHeatMonitor, RiskManager, and PaperEngine
- Integrate into the existing execution pipeline: every one of the 15 pre-trade checks must call event_logger.log_event() — passes with details of what was checked and what value was found, failures with the reason

MODULE 2: src/api/sse_streamer.py — SSEStreamer class

- Implements the /api/stream endpoint using Flask's Response with stream_with_context
- Pushes a JSON payload every 2 seconds with: account_info (equity, balance, daily_pnl), open_positions (list), portfolio_heat, vrs_scalar, rdd_statuses, recent_events (last 20 event log entries), active_signals (signals currently in pipeline)
- In PAPER mode: account_info comes from paper_account table; positions from paper_positions

- In DEMO/LIVE mode: account_info comes from mt5.account_info(); positions from mt5.positions_get()
- SSE format: data: {json_payload}\n\n — standard SSE format
- Client reconnection: the EventSource API handles this automatically; server must not hold state between connections

MODULE 3: src/paper/paper_engine.py — PaperEngine class

- __init__(self, config, db_manager, event_logger, mt5_connector=None)
- open_paper_position(symbol, direction, lot_size, entry_price, stop_loss, take_profit, signal_score, kelly_risk_pct, vrs_scalar, regime) → paper_ticket string
- close_paper_position(ticket, close_reason, close_price) → realized_pnl float
- update_all_positions(current_ohlc: dict) — called on every H1 bar close. Checks SL/TP/trailing stops/time exits against current bar data.
- simulate_fill_price(symbol, direction, requested_price) → float — applies slippage model from config
- get_open_positions() → list of position dicts with floating P&L calculated from current price
- reset_paper_account(archive=True) — archives current paper trade history with reset timestamp, resets balance

MODULE 4: src/api/chart_data.py — ChartDataProvider class

- get_ohlc_for_chart(symbol, timeframe, count=500) → list of {time, open, high, low, close} dicts — formatted for Lightweight Charts (Unix timestamps in seconds)
- get_indicators_for_chart(symbol, timeframe) → dict with keys: ema20, ema50, ema200, bb_upper, bb_mid, bb_lower — each a list of {time, value} dicts
- get_signals_for_chart(symbol, hours=168) → list of {time, direction, score, action} dicts for plotting arrows
- get_positions_for_chart(symbol) → list of {entry_price, sl, tp, direction, ticket} for horizontal lines

MODULE 5: src/analysis/readiness_checker.py — ReadinessChecker class

- evaluate() → list of dicts, one per condition: {condition, requirement, current_value, is_met: bool}
- Conditions to evaluate: demo runtime weeks, total trades per pair, win rate, profit factor, max drawdown, demo vs backtest deviation (calls comparison API), BQS consecutive weeks above 72, all critical audit gaps resolved (reads from a config list of known gaps), RDD status all pairs, VPS crash count from health monitor
- overall_ready() → bool — True only if ALL conditions return is_met=True
- Log each evaluation run to readiness_log table

MODULE 6: templates/index.html — Full Dashboard

- Implement all 7 zones exactly as specified in PRD Vol III Section 4
- Use CSS Grid for the main layout: header (fixed top), status strip (below header), then a 3-column grid: left sidebar (280px) | main panel (flex-grow) | right sidebar (260px), then positions panel (fixed bottom), then event log (collapsible bottom strip)
- All 6 tabs in Zone 4 must be implemented. Tabs switch content without a page reload.

- Connect to the SSE stream via EventSource('/api/stream'). On each message, update: account display in header, status strip KPI tiles, positions panel, right sidebar active signals, event log Zone 7.
- The chart tab loads via GET /api/v3/chart-data/<symbol>/<tf>. Redraw on symbol or timeframe change.
- The signals tab loads via GET /api/signals with filter support. Expand rows via GET /api/v3/signal-detail/<signal_id>.
- Dark trading theme: background #0D1117, panels #161B22, text #E6EDF3, green #3FB950, red #F85149, gold #D29922, blue #58A6FF.
- Mode indicator: read mode from GET /api/v3/mode on page load. Apply appropriate banner color and all mode-specific UI differences (paper vs demo vs live).

PART 5 — INTEGRATION REQUIREMENTS

After building each module, integrate it into the existing system:

1. Add event_log migration to database.py DatabaseManager.__init__()
2. Instantiate EventLogger in main.py and pass to all 15 pre-trade check callers
3. Add 15 event_logger.log_event() calls to the pre-trade check sequence (one per check step)
4. Instantiate PaperEngine in main.py. When mode=PAPER, replace OrderRouter.place_order() call with PaperEngine.open_paper_position()
5. Instantiate SSEStreamer and register /api/stream endpoint in Flask
6. Add all new Vol III API endpoints to src/api/main.py following existing Blueprint pattern
7. Add readiness_checker APScheduler job: runs every Friday 22:00 UTC alongside the existing Friday jobs
8. Wire paper position monitoring into the main H1 bar close loop: call paper_engine.update_all_positions(current_ohlc) on every bar

PART 6 — SESSION ORDER (IMPLEMENT IN THIS SEQUENCE)

Session	Module(s)	Why This Order
Session 1	event_log.py + integrate 15 log_event() calls into pre-trade pipeline	Foundation — every subsequent module depends on the event log
Session 2	sse_streamer.py + register /api/stream endpoint	Must be operational before the frontend can show live data
Session 3	paper_engine.py + paper_account.py + DB migrations	Paper engine before frontend so it can be tested independently
Session 4	chart_data.py + 3 new chart API endpoints	Frontend chart tab needs this API before it can render
Session 5	readiness_checker.py + /api/v3/readiness endpoint	Standalone module — can be built and tested in isolation
Session 6	All remaining /api/v3/ endpoints in src/api/main.py	Complete the API surface before building the frontend
Session 7	templates/index.html — Zones 1, 2, 3 + SSE connection + Tab 1 (Chart)	Build frontend in zone order; chart tab first as it's the most complex

Session	Module(s)	Why This Order
Session 8	Tabs 2 (Signals), 3 (Performance), 4 (Risk Dashboard)	Remaining tabs — all data APIs are now ready
Session 9	Tab 5 (Execution), Tab 6 (Weekly Report) + Zone 5 + Zone 6 + Zone 7	Complete frontend + positions panel + event log zone
Session 10	Full integration test on paper mode — all 6 symbols scanning, signals firing, paper positions opening	Validate everything works together end-to-end
Session 11	Switch to DEMO mode — full demo trading session, verify all demo-mode differences in UI	Final validation before live

═══ END OF APEX FX VOLUME III CODING PROMPT ═══

APPENDIX — Volume III Summary & Outstanding Items

9. Outstanding Items Before Live Trading

Based on the full audit in Part A, these items must be resolved before transitioning from DEMO to LIVE mode:

9.1 Critical Gaps (Fix Before Live)

Gap	Source	Fix Required
Trailing stop not called in main loop	Vol I / Master Prompt #1	Add position monitoring loop calling should_activate_trailing_stop() on every H1 bar
Symbol name normalization map	Master Prompt #2	Add SYMBOL_MAP to MT5Connector; validate all 6 symbols at startup
Position sizing formula discrepancy	Master Prompt #3	Audit calculate_position_size() — SL distance must be ATR×multiplier in price units
Spread-to-ATR filter missing	Vol II / Master Prompt #4	Add check_spread_atr_ratio() to OrderRouter — block if >0.15
Per-pair daily loss tracking	Master Prompt #5	Add get_daily_realised_pnl(symbol) to database.py; enforce in check_trade_allowed()
3-attempt retry (needs 4)	Vol II / Master Prompt #6	Rewrite retry loop to exactly 4 attempts with correct ms delays and slippage tolerances
No time-based trade exits	Master Prompt #7	Wire time_exit_manager.py into main monitoring loop
No weekend position policy	Master Prompt #8	Add friday_close_check() to time_exit_manager.py

9.2 Volume III Items to Build

Module	Priority	Session
event_log.py + 15-step pipeline logging	CRITICAL — needed for visibility	Session 1
sse_streamer.py — SSE real-time feed	CRITICAL — dashboard foundation	Session 2
paper_engine.py — virtual trading engine	HIGH — paper validation required	Session 3
chart_data.py — chart API	HIGH — primary interface	Session 4
readiness_checker.py — live readiness gate	MEDIUM — governance	Session 5

Module	Priority	Session
All /api/v3/ endpoints	HIGH — dashboard needs these	Session 6
Full dashboard index.html	CRITICAL — operator interface	Sessions 7-9
Integration tests for all V3 modules	HIGH — production quality	Session 10
DEMO mode full validation	CRITICAL — before live	Session 11

9.3 Volume III Success Criteria

Volume III is Complete When:

1. Dashboard loads in browser and shows correct mode banner (PAPER/DEMO/LIVE)2. All 6 instruments appear in the left sidebar with live regime badges3. The chart renders OHLC candles with EMA/BB overlays and signal arrows4. The SSE stream updates the header equity display and positions in real time5. In PAPER mode: signals fire, paper positions open, paper P&L tracks correctly across restarts6. In DEMO mode: real MT5 orders are placed, the Event Log shows all 15 pre-trade steps, fill latency is displayed7. The 'Ready for Live?' checklist correctly evaluates all 11 conditions8. Event log shows every pipeline decision with human-readable explanations9. All new API endpoints return correct data with proper error handling10. All tests pass (target: 160+ total tests including the new V3 test files)

APEX FX TRADING BOT — PRD Volume III

Frontend, Paper Trading & Demo Trading Visibility

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